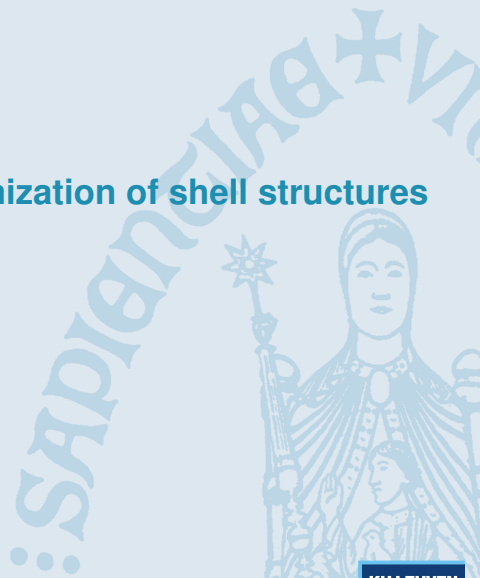
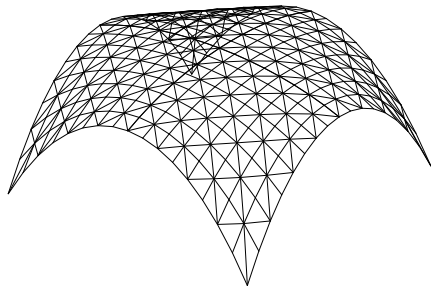


Chapter 7 – Shape optimization of shell structures

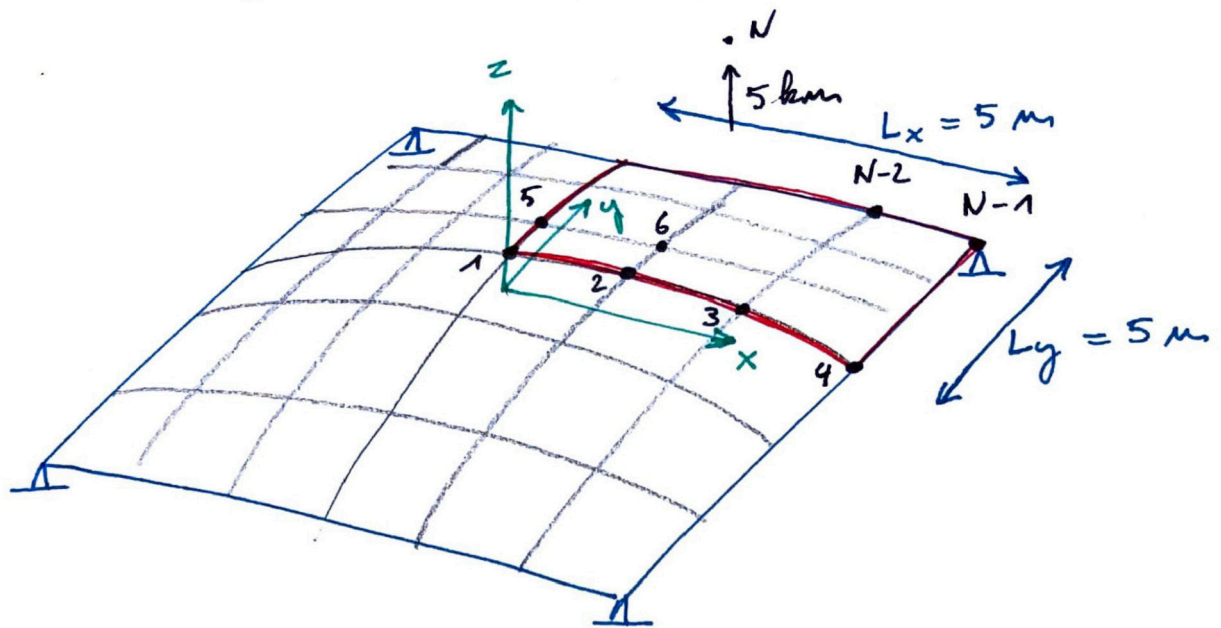


Example: size and shape optimization of a gridshell



- Width: 10 m.
- Depth: 10 m.
- Corners are pin-supported.
- Vertical distributed load of 1 kN/m^2 per horizontal unit of area.
- All members have a square cross section with side length h .
- The orthogonal members are beam elements, the diagonal members are bar elements.
- Young's modulus $E_L = 10 \text{ GPa}$, shear moduli $\mu_{TR} = \mu_{RT} = 0.098 \text{ GPa}$.
- The objective is to minimize compliance subject to a volume constraint $V_{\max} = 0.8 \text{ m}^3$.
- Design variables: all unsupported nodes can move vertically up to a height of 10 m; the side length h of the member's cross section can vary between 8 mm and 200 mm.

Geometry and boundary conditions



FE model

$$\text{Nodes} = \begin{bmatrix} i_{m1} & x_{m1} & y_{m1} & \boxed{z_{m1}} \\ \vdots & \vdots & \vdots & \vdots \\ i_{mN-2} & x_{mN-2} & y_{mN-2} & z_{mN-2} \\ i_{mN-1} & x_{mN-1} & y_{mN-1} & z_{mN-1} \\ i_{mN} & x_{mN} & y_{mN} & z_{mN} \end{bmatrix}$$

$$\text{Sections} = \begin{bmatrix} 1 & h^2 & \infty & \infty & \beta h^4 & h^4/12 & h^4/12 \\ 2 & h^2/2 & \infty & \infty & \beta h^4/2 & h^4/24 & h^4/24 \end{bmatrix}$$

Torsional constant I_{xx} for rectangular sections: $I_{xx} = \beta h b^3$. For square sections, $b = h$ and $\beta = 0.1406$.

$$\text{Materials} = \begin{bmatrix} 1 & E & \nu \end{bmatrix}$$

For Euler beams, ν is only used to compute the shear modulus $\mu = \frac{E}{2(1+\nu)}$ to determine torsion stiffness

→ we take $\nu = 50$ to simulate an orthotropic material.

Parameterization

$$\text{Design variables } x = \begin{bmatrix} z_{m1} \\ \vdots \\ z_{mN-2} \\ h \end{bmatrix}$$

The modal z -coordinates z_m and the bar size h are obtained as:

$$z_m = A_m x$$

$$h = A_h x$$

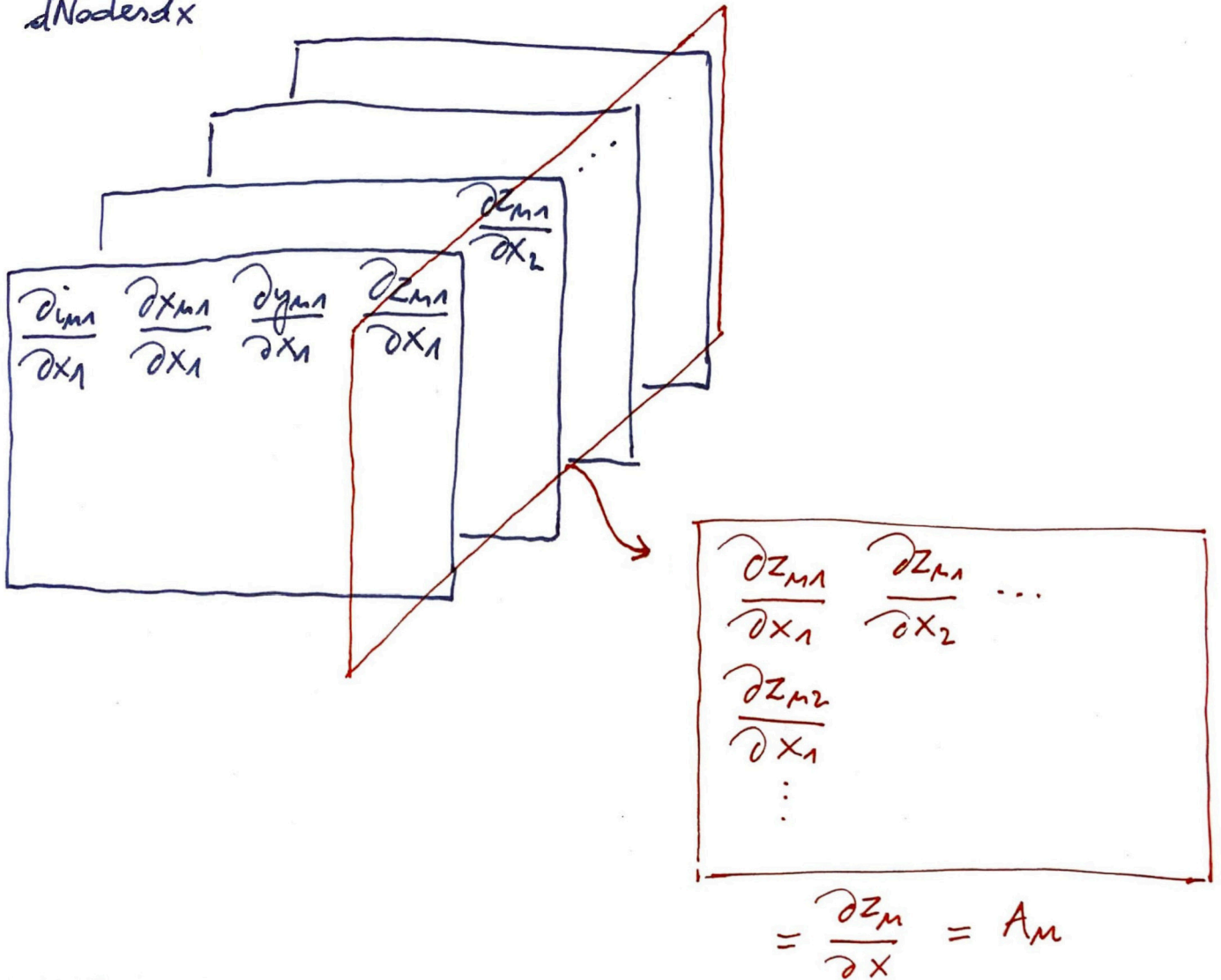
where:

$$A_m = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 & \\ & & & & & 0 \end{bmatrix}$$

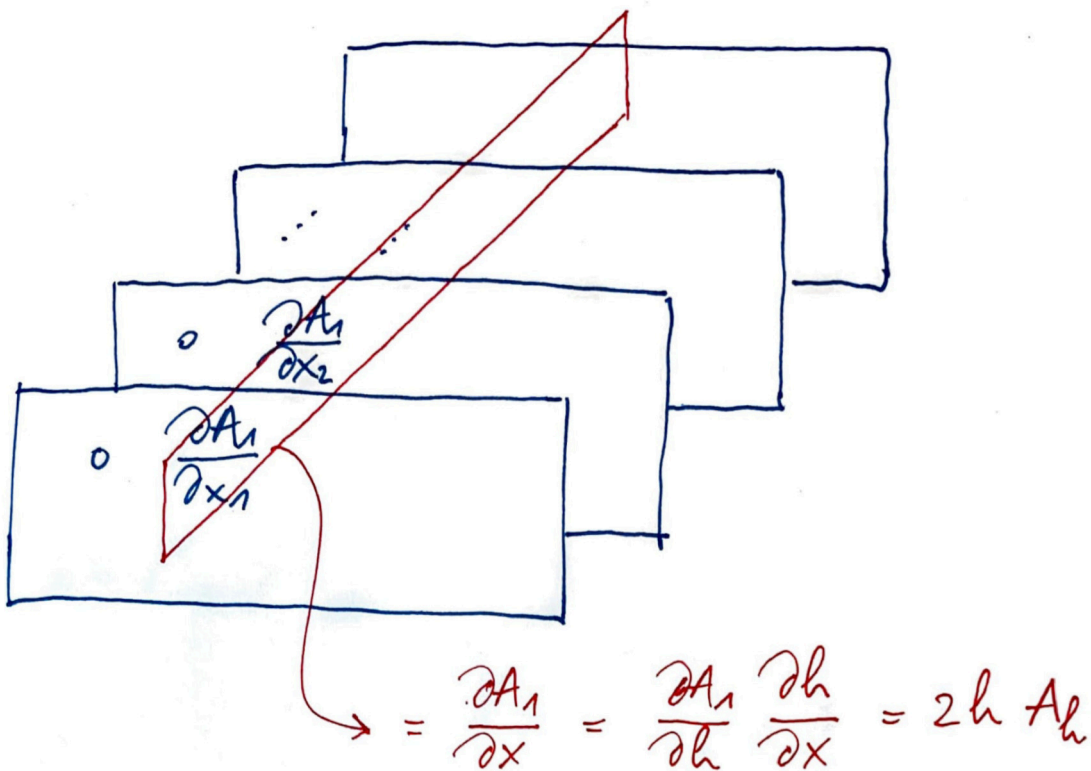
$$A_h = [0 \ 0 \ 0 \ \dots \ 0 \ 1]$$

Sensitivities

$dNodes dx$



$dSections dx$



Example: size and shape optimization of a gridshell

MATLAB code (essential lines only; see also ex1.m):

```
% INITIALIZE DESIGN VARIABLES
x = [3.333; 3.333; 3.333; ...; 3.333; 0.04]; % Initial design vector

% OPTIMIZATION LOOP
while true

    % UPDATE NODES MATRIX
    zn = An*x; % Vertical node coordinates
    Nodes(1:nNode-1,4) = zn; % Nodes matrix

    % UPDATE SECTIONS MATRIX
    h = Ah*x; % Bar section side length
    Sections(:, [2 5 6 7]) = [h^2 beta*h^4 h^4/12 h^4/12;
                             h^2/2 beta*h^4/2 h^4/24 h^4/24];

    % NODE DEFINITION DERIVATIVES
    dzndx = An; % Derivative of zn w.r.t. x
    dNodesdx = zeros([size(Nodes), n]); % Derivative of Nodes w.r.t. x
    dNodesdx(1:nNode-1,4,:) = dzndx;

    % SECTION DEFINITION DERIVATIVES
    dSectionsdx = zeros([size(Sections), n]); % Derivative of Sections w.r.t. x
    dSectionsdx(1,2,:) = 2*h*Ah;
    ...
end
```

Splines

- A spline $[x(u), y(u), z(u)]$ in 3D space is a vector-valued parametric function of a parameter $u \in [0, 1]$.

$$x(u) = \sum_{i=1}^N X_i B_{i,p}(u) = \mathbf{B}_p(u) \mathbf{X}$$

$$y(u) = \sum_{i=1}^N Y_i B_{i,p}(u) = \mathbf{B}_p(u) \mathbf{Y}$$

$$z(u) = \sum_{i=1}^N Z_i B_{i,p}(u) = \mathbf{B}_p(u) \mathbf{Z}$$

- The x -, y -, and z -components of a spline are defined as linear combinations of N piecewise polynomial basis functions $B_{i,p}(u)$ of degree p with coefficients X_i , Y_i , and Z_i .
- The coefficients X_i , Y_i , and Z_i can be regarded as points in space; these points $\mathbf{V}_i = [X_i, Y_i, Z_i]$ are the so-called control vertices of the spline.
- Splines of degree p are $p - 1$ times continuously differentiable.

Splines

- The position in space of a point $[x(u^*), y(u^*), z(u^*)]$ of the spline (corresponding to a specific parameter value u^*) depends linearly on the positions $[X_i, Y_i, Z_i]$ of the control vertices $\mathbf{V}_i$:

$$\frac{\partial x(u^*)}{\partial X_i} = B_{i,p}(u^*)$$

$$\frac{\partial y(u^*)}{\partial Y_i} = B_{i,p}(u^*)$$

$$\frac{\partial z(u^*)}{\partial Z_i} = B_{i,p}(u^*)$$

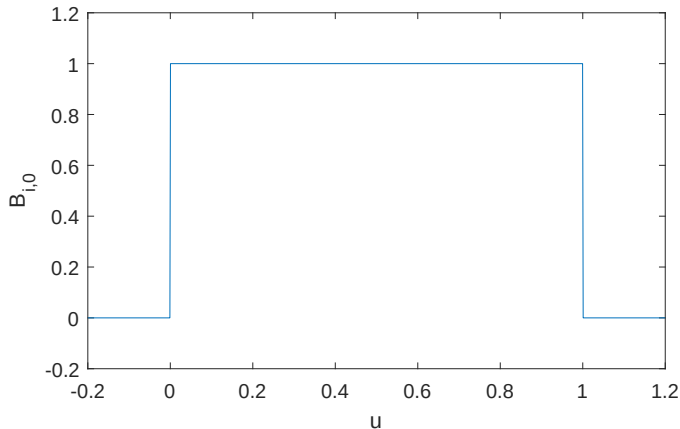
$$\frac{\partial x(u^*)}{\partial \mathbf{X}} = \mathbf{B}_p(u^*)$$

$$\frac{\partial y(u^*)}{\partial \mathbf{Y}} = \mathbf{B}_p(u^*)$$

$$\frac{\partial z(u^*)}{\partial \mathbf{Z}} = \mathbf{B}_p(u^*)$$

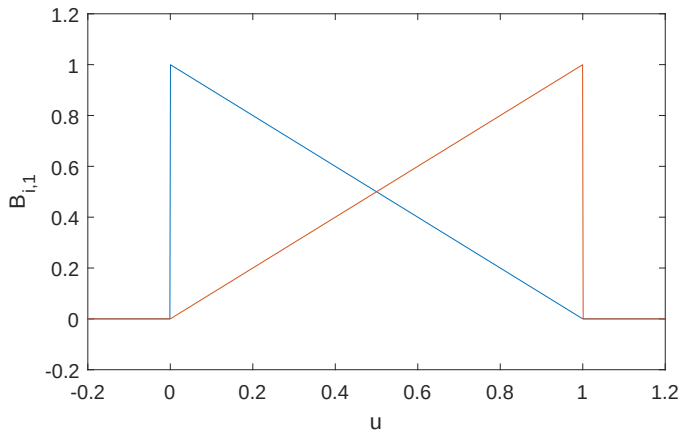
Bezier spline basis functions

- $N = 1$ Bezier spline basis function of degree $p = 0$ (see also `ex2a.m`):



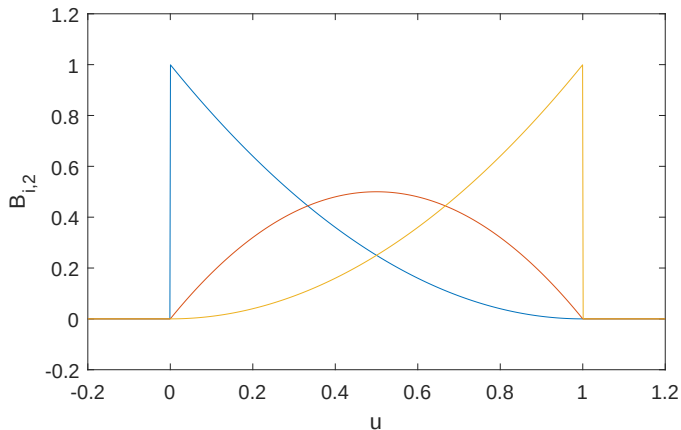
Bezier spline basis functions

- $N = 2$ Bezier spline basis functions of degree $p = 1$ (see also `ex2a.m`):



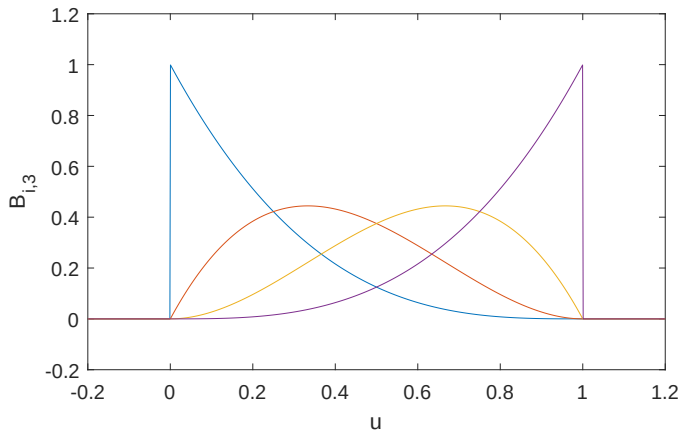
Bezier spline basis functions

- $N = 3$ Bezier spline basis functions of degree $p = 2$ (see also `ex2a.m`):



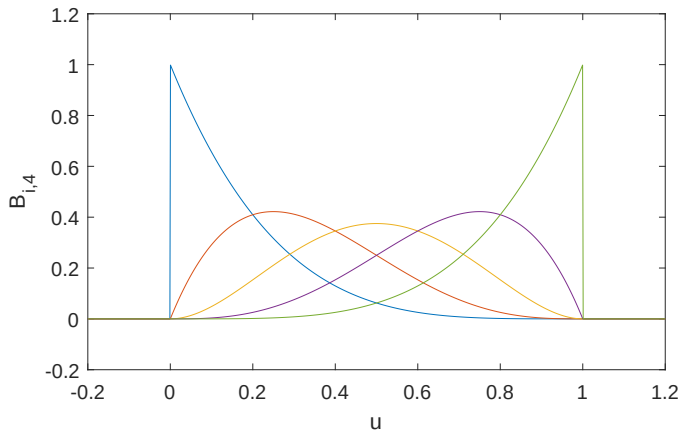
Bezier spline basis functions

- $N = 4$ Bezier spline basis functions of degree $p = 3$ (see also `ex2a.m`):



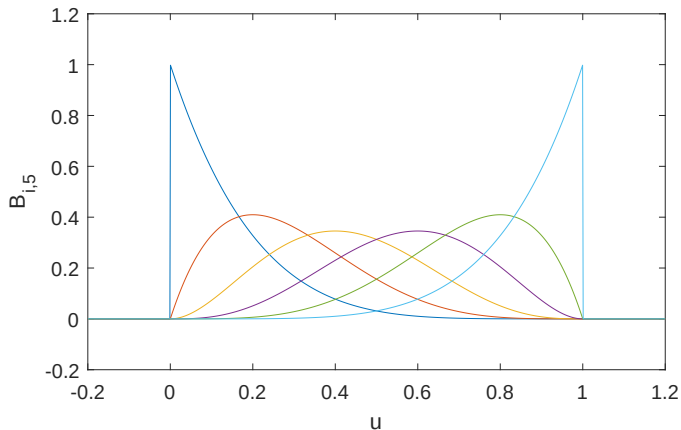
Bezier spline basis functions

- $N = 5$ Bezier spline basis functions of degree $p = 4$ (see also `ex2a.m`):



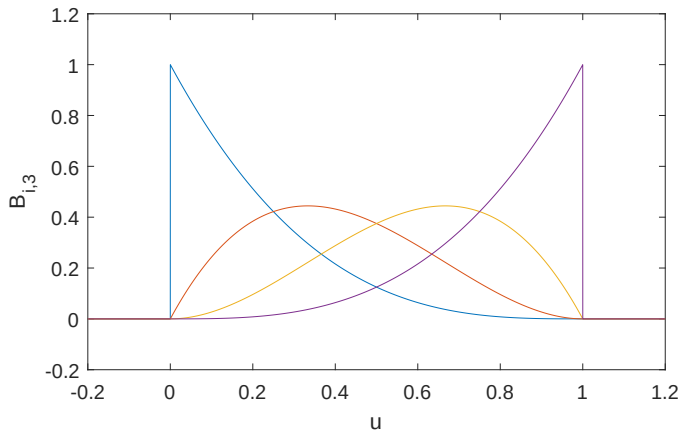
Bezier spline basis functions

- $N = 6$ Bezier spline basis functions of degree $p = 5$ (see also `ex2a.m`):



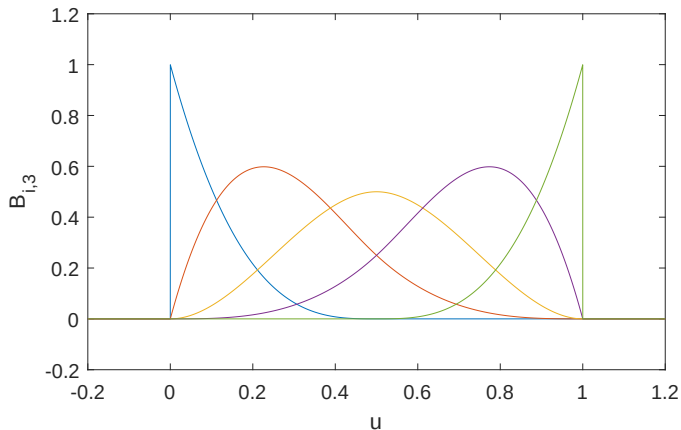
B-spline basis functions

- $N = 4$ B-spline basis functions of degree $p = 3$ (see also ex2b.m):



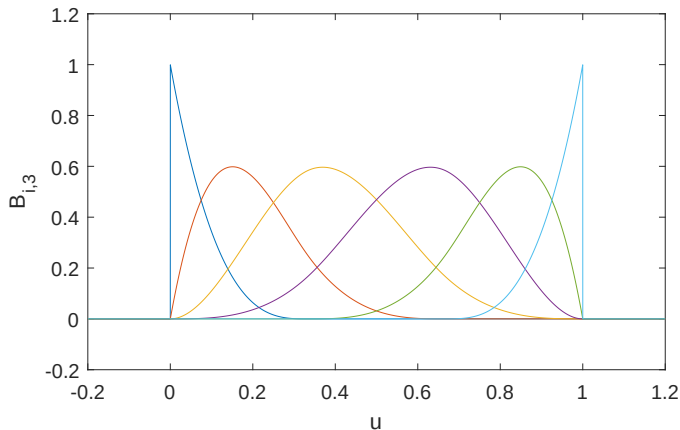
B-spline basis functions

- $N = 5$ B-spline basis functions of degree $p = 3$ (see also ex2b.m):



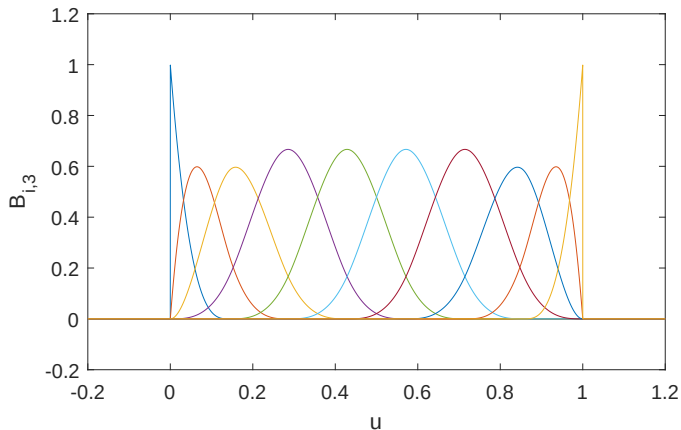
B-spline basis functions

- $N = 6$ B-spline basis functions of degree $p = 3$ (see also `ex2b.m`):



B-spline basis functions

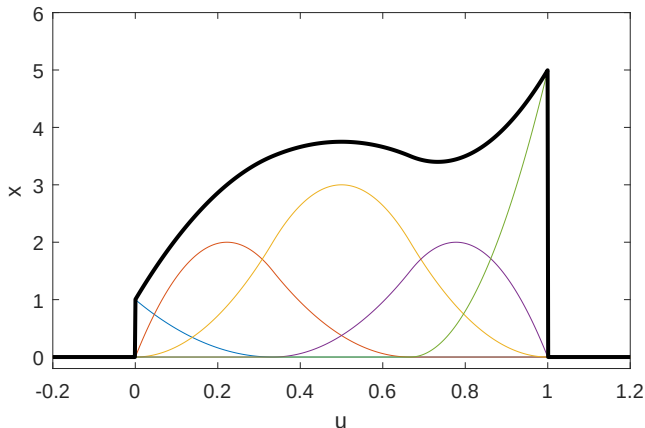
- $N = 10$ B-spline basis functions of degree $p = 3$ (see also ex2b.m):



B-spline examples

- 2nd degree B-spline $x(u)$ with 5 control vertices (see also ex3.m):

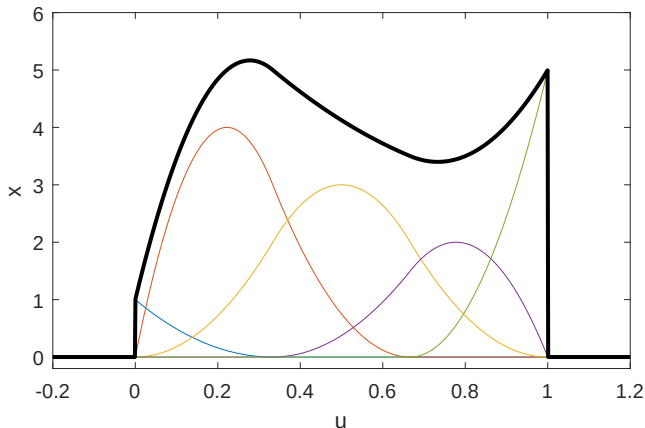
$$X_1 = 1, \quad X_2 = 3, \quad X_3 = 4, \quad X_4 = 3, \quad X_5 = 5$$



B-spline examples

- 2nd degree B-spline $x(u)$ with 5 control vertices (see also ex3.m):

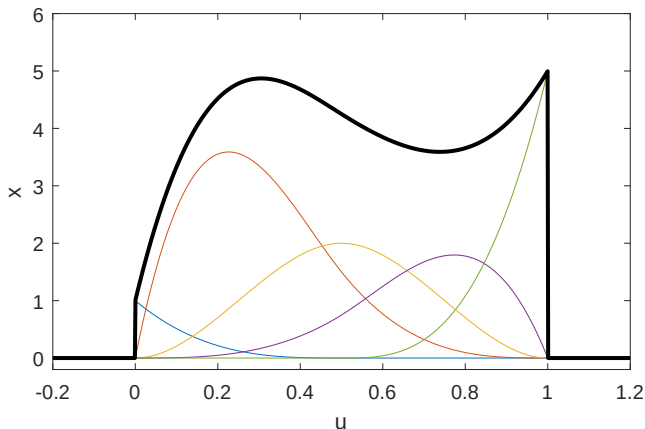
$$X_1 = 1, \quad X_2 = 6, \quad X_3 = 4, \quad X_4 = 3, \quad X_5 = 5$$



B-spline examples

- 3rd degree B-spline $x(u)$ with 5 control vertices (see also ex3.m):

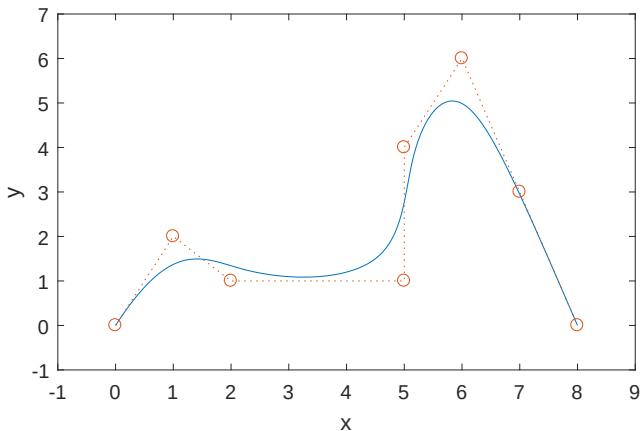
$$X_1 = 1, \quad X_2 = 6, \quad X_3 = 4, \quad X_4 = 3, \quad X_5 = 5$$



B-spline examples

- 3rd degree B-spline $[x(u), y(u)]$ with 8 control vertices:

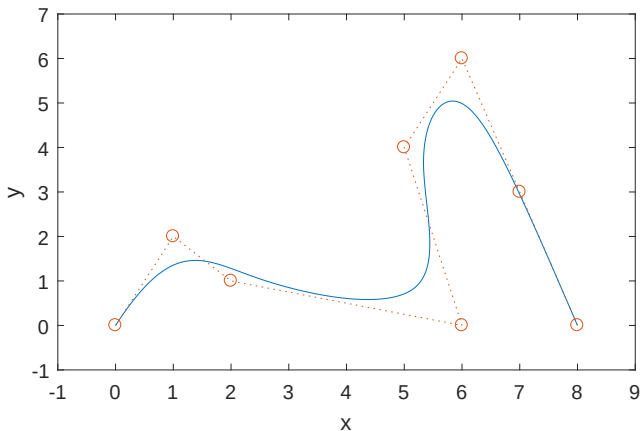
$$\mathbf{V}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{V}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{V}_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{V}_4 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \mathbf{V}_5 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \mathbf{V}_6 = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \mathbf{V}_7 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}, \mathbf{V}_8 = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$



B-spline examples

- 3rd degree B-spline $[x(u), y(u)]$ with 8 control vertices:

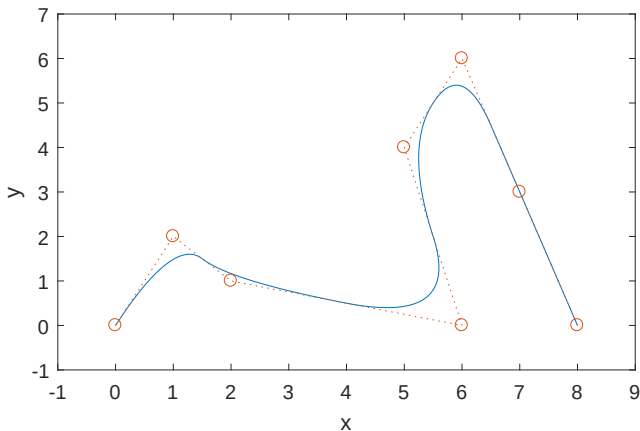
$$\mathbf{V}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{V}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{V}_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{V}_4 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \mathbf{V}_5 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \mathbf{V}_6 = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \mathbf{V}_7 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}, \mathbf{V}_8 = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$



B-spline examples

- 2nd degree B-spline $[x(u), y(u)]$ with 8 control vertices:

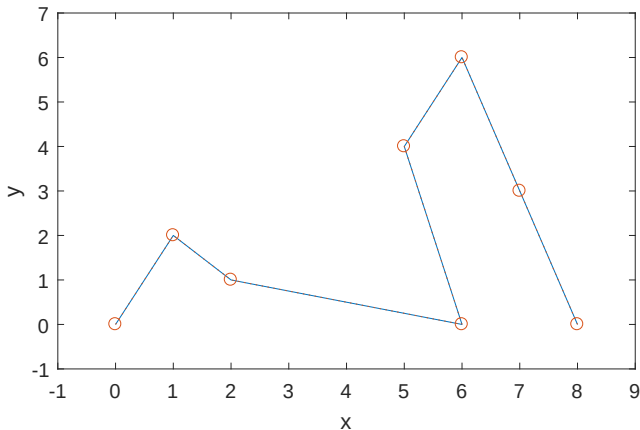
$$\mathbf{V}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{V}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{V}_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{V}_4 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \mathbf{V}_5 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \mathbf{V}_6 = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \mathbf{V}_7 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}, \mathbf{V}_8 = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$



B-spline examples

- 1st degree B-spline $[x(u), y(u)]$ with 8 control vertices:

$$\mathbf{V}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{V}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{V}_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{V}_4 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \mathbf{V}_5 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \mathbf{V}_6 = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \mathbf{V}_7 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}, \mathbf{V}_8 = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$



B-spline examples

MATLAB code for the B-spline example (see also ex4.m):

```
% B-SPLINE BASIS FUNCTIONS
p = 3;                                % Spline degree
N = 8;                                % Number of control vertices
u = linspace(0,1,5000);               % Free parameter sampling
B = bspline(p,N,u);                   % B-spline basis functions evaluated at u

% CONTROL VERTICES
X = [0 1 2 5 5 6 7 8]';               % Control vertex x-coordinates
Y = [0 2 1 1 4 6 3 0]';               % Control vertex y-coordinates

% GENERATE SPLINE
x = B*X;                              % Spline point x-coordinates
y = B*Y;                              % Spline point y-coordinates

% DERIVATIVES
dxdX = B;                             % Derivative of x w.r.t. X
dydY = B;                             % Derivative of y w.r.t. Y

% PLOT SPLINE
figure;
plot(x,y,X,Y,'o');
```

Spline surfaces

- A spline surface $[x(u, v), y(u, v), z(u, v)]$ in 3D space is a vector-valued parametric function of two parameters $u \in [0, 1]$ and $v \in [0, 1]$:

$$x(u, v) = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} X_{ij} B_{i,p_u}(u) B_{j,p_v}(v) = \mathbf{B}_{p_u}(u) \mathbf{X} \mathbf{B}_{p_v}^T(v)$$

$$y(u, v) = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} Y_{ij} B_{i,p_u}(u) B_{j,p_v}(v) = \mathbf{B}_{p_u}(u) \mathbf{Y} \mathbf{B}_{p_v}^T(v)$$

$$z(u, v) = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} Z_{ij} B_{i,p_u}(u) B_{j,p_v}(v) = \mathbf{B}_{p_u}(u) \mathbf{Z} \mathbf{B}_{p_v}^T(v)$$

- The x -, y -, and z -components of a spline surface are defined as linear combinations of $N_u N_v$ products of spline basis functions $B_{i,p_u}(u) B_{j,p_v}(v)$ with coefficients X_{ij} , Y_{ij} , and Z_{ij} .
- The coefficients X_{ij} , Y_{ij} , and Z_{ij} can be regarded as points in space; these points $\mathbf{V}_{ij} = [X_{ij}, Y_{ij}, Z_{ij}]$ are the so-called control vertices of the spline surface.

Spline surfaces

- The position in space of a point $[x(u^*, v^*), y(u^*, v^*), z(u^*, v^*)]$ of the spline surface (corresponding to specific parameter values u^* and v^*) depends linearly on the positions $[X_{ij}, Y_{ij}, Z_{ij}]$ of the control vertices $\mathbf{V}_{ij}$:

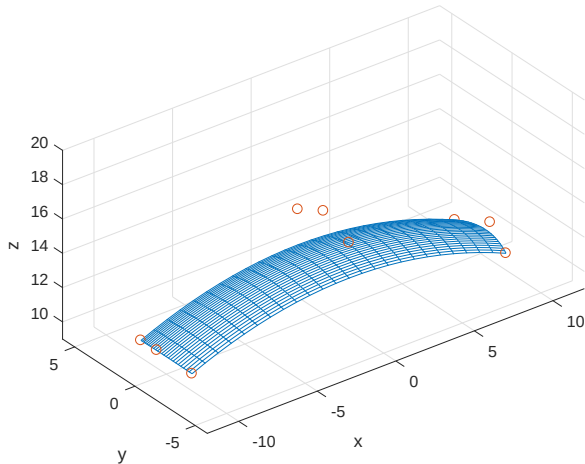
$$\frac{\partial x(u^*, v^*)}{\partial X_{ij}} = B_{i,p_u}(u^*) B_{j,p_v}(v^*)$$

$$\frac{\partial y(u^*, v^*)}{\partial Y_{ij}} = B_{i,p_u}(u^*) B_{j,p_v}(v^*)$$

$$\frac{\partial z(u^*, v^*)}{\partial Z_{ij}} = B_{i,p_u}(u^*) B_{j,p_v}(v^*)$$

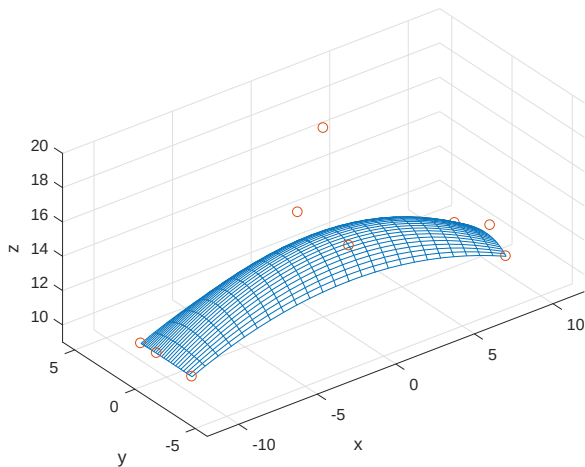
B-spline surface example

- 2nd degree B-spline surface $[x(u, v), y(u, v), z(u, v)]$ with 3×3 control vertices:



B-spline surface example

- 2nd degree B-spline surface $[x(u, v), y(u, v), z(u, v)]$ with 3×3 control vertices; now the center vertex has been moved upwards:



Spline surfaces

MATLAB code for the spline surface example (see also `ex5.m`):

```
% B-SPLINE BASIS FUNCTIONS
pu = 2;           % Spline degree in terms of u
pv = 2;           % spline degree in terms of v
Nu = 3;           % Number of control vertices in u-direction
Nv = 3;           % Number of control vertices in v-direction
u = linspace(0,1,25); % Free parameter sampling in u-direction
v = linspace(0,1,25); % Free parameter sampling in v-direction
Bu = bspline(pu,Nu,u); % B-spline basis functions in u-direction
Bv = bspline(pv,Nv,v); % B-spline basis functions in v-direction

% CONTROL VERTICES
X = [ 9.99  0.00 -9.99; % Control vertex x-coordinates
      10.61  0.00 -10.61;
      9.99  0.00 -9.99];
Y = [-2.12 -2.12 -2.12; % Control vertex y-coordinates
      0.00  0.00  0.00;
      2.12  2.12  2.12];
Z = [ 9.99 14.12  9.99; % Control vertex z-coordinates
      10.61 15.00 10.61;
      9.99 14.12  9.99];
```

Spline surfaces

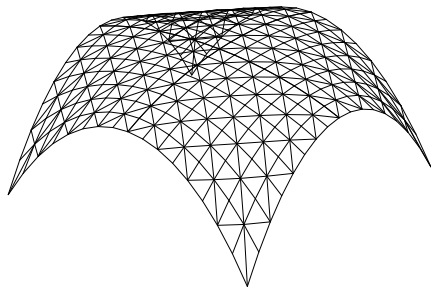
MATLAB code for the spline surface example, continued:

```
% GENERATE SPLINE SURFACE
x = Bu*X*Bv';           % Spline surface point x-coordinates
y = Bu*Y*Bv';           % Spline surface point y-coordinates
z = Bu*Z*Bv';           % Spline surface point z-coordinates

% DERIVATIVES
dxdX = kron(Bv,Bu);      % Derivative of x(:) w.r.t. X(:)
dydY = kron(Bv,Bu);      % Derivative of y(:) w.r.t. Y(:)
dzdZ = kron(Bv,Bu);      % Derivative of z(:) w.r.t. Z(:)

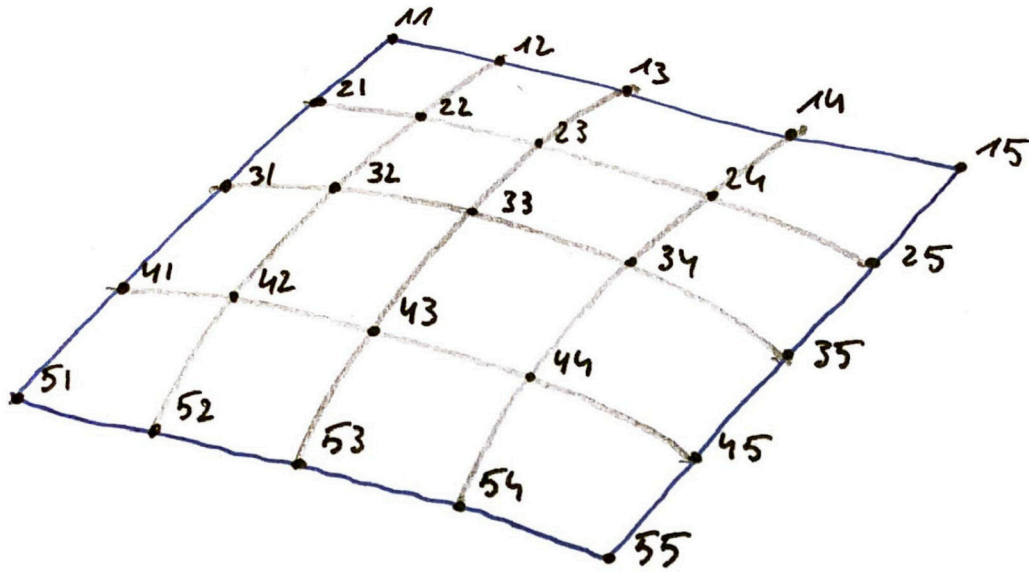
% PLOT SPLINE SURFACE
figure;
mesh(x,y,z);
hold('on');
plot3(X(:),Y(:),Z(:),'o');
```


Example: size and shape optimization of a gridshell



- Solve the size and shape optimization problem introduced on slide 110 using a symmetric-symmetric 4th degree Bezier spline surface to represent the shape of the shell (ex6a.m).

Spline surface: vertices and parameterization



z-coordinates of the vertices:

$$Z_s = \begin{bmatrix} Z_{s11} & Z_{s12} & Z_{s13} & Z_{s14} & Z_{s15} \\ Z_{s21} \\ \vdots \end{bmatrix}$$

These z-coordinates are controlled by 8 design variables:

$$Z_s = \begin{bmatrix} 0 & x_3 & x_6 & x_3 & 0 \\ x_1 & x_4 & x_7 & x_4 & x_1 \\ x_2 & x_5 & x_8 & x_5 & x_2 \\ x_1 & x_4 & x_7 & x_4 & x_1 \\ 0 & x_3 & x_6 & x_3 & 0 \end{bmatrix}$$

or, in vector format:

$$Z_s = \begin{bmatrix} Z_{s11} \\ Z_{s21} \\ \vdots \\ Z_{s12} \\ Z_{s22} \\ \vdots \end{bmatrix}$$

These coordinates are obtained from the design variables x as:

$$Z_s = A_s \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_3 \\ x_3 \end{bmatrix} \quad \text{bar size}$$

Nodal coordinates

$$Z_m = (B_u Z_s B_v^T)$$

evaluate for u from 0.5 to 1
 v from 0.5 to 1

or, in vector format:

$$Z_m = \begin{bmatrix} Z_{m11} \\ Z_{m21} \\ \vdots \end{bmatrix}$$

Sensitivities

$$\frac{\partial Z_m}{\partial x} = \frac{\partial Z_m}{\partial Z_s} \frac{\partial Z_s}{\partial x}$$

$$L = A_s$$

$$L \text{ in Matlab: } = \text{ kron}(B_v, B_u)$$

Example: size and shape optimization of a gridshell

MATLAB code (essential lines only):

```
% INITIALIZE DESIGN VARIABLES
x = [3.333; 3.333; 3.333; ...; 3.333; 0.04]; % Initial design vector

% PARAMETRIC SHAPE REPRESENTATION
p = 4; % Spline degree
u = linspace(0.5,1,nx+1); % u-values corresponding to FE nodes
v = linspace(0.5,1,ny+1); % v-values corresponding to FE nodes
Bu = bspline(p,[],u); % Spline basis functions in u-dimension
Bv = bspline(p,[],v); % Spline basis functions in v-dimension
```

Example: size and shape optimization of a gridshell

MATLAB code (essential lines only), continued:

```
% OPTIMIZATION LOOP
while true

    % UPDATE NODES MATRIX
    zs = As*x;
    Zs = reshape(zs,Nu,Nv);
    Zn = Bu*Zs*Bv';
    zn = Zn(:);
    Nodes(1:nNode-1,4) = zn;

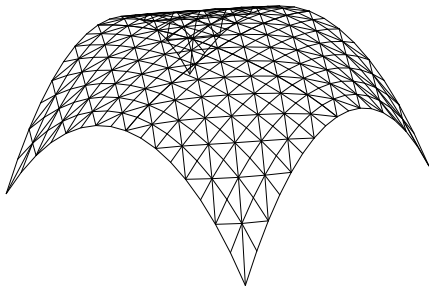
    % Control vertex z-coordinates
    % Idem in matrix format
    % Nodal z-coordinates in matrix format
    % Idem in vector format
    % Nodes matrix

    % NODE DEFINITION DERIVATIVES
    dzndzs = kron(Bv,Bu);
    dzsdx = As;
    dzndx = dzndzs*dzsdx;
    dNodesdx = zeros([size(Nodes),n]);
    dNodesdx(1:nNode-1,4,:) = dzndx;

    % Derivative of zn=Zn(:) w.r.t. zs=Zs(:)
    % Derivative of zs w.r.t. x
    % Derivative of zn w.r.t. x (via chain rule)
    % Derivative of Nodes w.r.t. x

end
```

Example: size and shape optimization of a gridshell



- Add extra constraints to ensure that the entrance height (nodes 9 and 73) is at least 5 m (ex6b.m).
- Keep the entrance height constraints and add an extra constraint to ensure that the crown height (node 1) does not exceed 7 m (ex6c.m).
- Compare the compliance of the optimized design for the three cases considered.

Example: size and shape optimization of a gridshell

Additional code needed to control the entrance and crown height:

```
% Z-COORDINATES OF NODES A, B AND C
zA = Nodes(1,4); % Z-coordinate of the center node
zB = Nodes(nx+1,4); % Z-coordinate of the east entrance node
zC = Nodes(ny*(nx+1)+1,4); % Z-coordinate of the north entrance node

% NODAL Z-COORDINATE SENSITIVITIES
dzAdx = reshape(dNodesdx(1,4,:),1,[]);
dzBdx = reshape(dNodesdx(nx+1,4,:),1,[]);
dzCdx = reshape(dNodesdx(ny*(nx+1)+1,4,:),1,[]);
```